

LABORATORY REPORT

Laboratory Activity 3: JavaScript Fundamentals 1

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Problem Sets

Problem 1: Grade Calculator

```
let base_number = 3; //global variable : base number used in all functions

// Problem 1: Grade Calculator
function calculateGrade(score) { //calculate letter grade base on score
  //conditional statements used to determine the letter grade
  if (score<=100 && score >= 90)
    return 'A';
  if (score >= 80)
    return 'B';
  if (score >= 70)
    return 'C';
  if (score >= 60)
    return 'D';
  return 'F';
}

let score = base_number * 10 + 5; //formula for score
//Printing
console.log("Problem 1: Grade Calculator");
console.log("Score = ", score, "\nGrade = ", calculateGrade(score)); //calls function
console.log("-----");
```

Description:

This program calculates the letter grade for a given score.

Variables:

1. **base_number** – a global variable set to 3 (last digit of my student ID)
2. **score** – its value will be based on **base_number**

Steps I did:

- Declared a global variable **base_number** and set it to 3.
- Created a function **calculateGrade(score)** to determine the letter grade based on **score**.
- Used if and else conditional statements to assign the correct letter grade according to the grading rules provided in the pdf.

- Declared a variable **score** outside the function and calculated it using the formula: **score = base_number × 10 + 5**.
- Printed the score and its corresponding letter grade.

OUTPUT

```
PS C:\Users\User> node "c:\Users\User\Desktop\ALAGBATE_LAB3.js"
Problem 1: Grade Calculator
Score = 35
Grade = F
-----
```

Problem 2: Star Pattern

```
// Problem 2: Star Pattern
function showStars(rows) { //function to print triangle of stars
  // Outer loop: controls the number of rows
  for (let row = 1; row <= rows; row++) {
    // Start with an empty string for each row
    let pattern = '';
    // Inner loop: controls how many stars are added per row
    for (let i = 0; i < row; i++)
      pattern += '*'; // Add one star to the pattern string

    //Printing
    console.log(pattern);
  }
}

let rows = base_number + 2; // formula for rows
//Printing
console.log("Problem 2: Star Pattern");
showStars(rows); //calls function
console.log("-----");
```

Description:

This program uses loops to print a triangle of stars.

Variables:

1. **base_number** – global variable set to 3 that will later on be used to set the value of the number of rows.
2. **rows** – number of rows in the triangle of stars (its value will be based on **base_number**)
3. **row** – loop counter for the outer loop controlling rows
4. **i** – loop counter for the inner loop controlling stars per row
5. **pattern** – stores the stars for the current row before printing

Steps I did:

- Declared a function **showStars(rows)** to print the triangle of stars.
- Used a nested loop: the outer loop controls the number of rows, and the inner loop adds the correct number of stars per row.
- Calculated the number of rows using **rows = base_number + 2**.
- Called the function to print the triangle of stars to the console.

OUTPUT

```
Problem 2: Star Pattern
*
**
***
****
*****
-----
```

Problem 3: Prime Number Checker

```
// Problem 3: Prime Number Checker
function isPrime(n) { //function to check if number is prime
  if (n < 2)
    return false; //numbers less than 2 are not prime
  //loop to determine if number is prime
  for (let factor = 2; factor <= Math.sqrt(n); factor++) { //start checking from 2
    if (n % factor === 0) return false; // If divisible by any factor, it's not prime
  }
  return true; //If no divisor was found then the number is prime
}

let n = base_number + 10; //formula for n
//printing
console.log("Problem 3: Prime Number Checker");
console.log("Is", n, "a prime number?");

if (isPrime(n)) {
  console.log("Prime"); //if function returns true, print Prime
} else {
  console.log("Not Prime"); //if function returns false, print Not Prime
}
console.log("-----");
```

Description:

This program checks whether a number is prime.

Variables:

1. **base_number** – used to calculate n

2. **n** – number to check (its value will be based on **base_number**)
3. **factor** – loop counter used to test divisibility

Steps I did:

- Declared a function **isPrime(n)** that determines if a number is prime.
- Checked if **n** is less than 2 and returned **false** if it is.
- Used a loop to test divisibility from 2 up to the square root of **n**; if any factor divides **n**, returned **false**.
- If no divisors were found, returned **true**.
- Calculated **n** using **n = base_number + 10**.
- Printed whether **n** is prime using an if-else statement to display "Prime" or "Not Prime" in the console.

OUTPUT

```
Problem 3: Prime Number Checker
Is 13 a prime number?
Prime
-----
```

Problem 4: Multiplication Table

```
// Problem 4: Multiplication Table
function multiplicationTable(n) { //function to print multiplication table of base number 3 from 1-10
  for (let i = 1; i <= 10; i++) {
    console.log(n + " x " + i + " = " + (n * i));
  }
}
//printing
console.log("Problem 4: Multiplication Table");
multiplicationTable(base_number); //calls function
```

Description:

This program prints the multiplication table of **n** from 1 to 10.

Variables:

1. **base_number** – global variable set to 3
2. **n** – equal to **base_number**

3. **i** – loop counter used to calculate products from 1 to 10

Steps I did:

- Declared a function **multiplicationTable(n)** to print the multiplication table.
- Used a loop from 1 to 10 to calculate and print each product ($n \times i$).
- Called the function with **base_number** to print its multiplication table to the console.

OUTPUT

```
Problem 4: Multiplication Table
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
```

Github repository: [LoidaAlagbate/ALAGBATE-APPDEV-LAB3](https://github.com/LoidaAlagbate/ALAGBATE-APPDEV-LAB3)

